

- Question paper is for 70 points; 120 minutes.

SECTION-I

1. (4 points) As a broad classification, organic reactions can be classified into four types; list, explain each in a sentence or two and give an example for each of them.
2. (6 points) (a) Consider the n-butane molecule. Draw schematic diagram of the torsional energy of this molecule for the torsion around $C_2 - C_3$ σ bond. (3pts) (b) Consider the cyclopentane molecule. Draw the stable configurations of this molecule and order them according to their stability. Comment. (3pts)
3. (6 points) Discuss the various effects of the solvent on/in the various organic reactions. Illustrate with an example each.
4. (6 points) We had discussed the nucleophilic substitutions at aliphatic carbons (specifically, S_N^1 and S_N^2 mechanisms). Identify the various factors that may effect the mechanism, and write 2/3 sentences on each of the factors
5. (6 points) What are kinetic and thermodynamic control of a reaction? Explain clearly the differences and origin of these differences. Give a real example/s which demonstrate the concept/s.
6. (4 points) Draw the structures of the two isomers of the molecule 1-chloro 2-isopropyl cyclohexane clearly indicating the axial and equatorial substitutions. Each isomers undergo elimination reaction (of HCl molecule). Indicate the products (and expected product ratio's). Which isomer has faster rate of elimination and why? (NOTE: not explicitly discussed in lectures, question is based on your prior Organic Chemistry exposure)
7. (6 points) (a) Explain NMR spectroscopy in about 5 sentences (2 points). (b) For ^{13}C and 1H spectra (NMR), list the range where signals are expected for 5 different "functional groups" (for each) (4 points)
8. (4 points) Answer the following for IR spectroscopy: List the regions of IR spectra for 8-10 "functional groups" or chemical moieties.
9. (6 points) What is aromaticity? For a molecule to be aromatic, what are the conditions? Using the energy level diagrams of molecular orbitals, illustrate aromaticity (for molecules having 4, 5, 6 and 8 carbon atoms), explain.
10. (4 points) Using an example, demonstrate how isotopes are used to determine the mechanism of reaction.
11. (6 points) Why does the nucleophile approach the carbonyl atom differ from perpendicular (90°) attack on $C = O$ bond? What is the actual angle of attack? Using MO's describe the various factors that determine this angle.
12. (6 points) Use of sodium bisulphite is a preferred, in comparison to direct route, to form cyanohydrins. Why? Give example and illustrate.
13. (6 points) (a) List the factors that determine the acidity/basicity of organic molecules (2 pts). (b) For aromatic compounds, how do substituents determine the effect the acidity? Give an example for each such cause. (4pts)

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